

Optimization of Mueller matrix polarimeters in the presence of error sources

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Abstract: Methods are presented for optimizing the design of Mueller matrix polarimeters and in particular selecting the retardances and orientation angles of polarization components to ensure accurate reconstruction of a sample's Mueller matrix in the presence of error sources. Metrics related to the condition number and to the singular value decomposition are used to guide the design process for Mueller matrix polarimeters with the goal of specifying polarization elements, comparing polarimeter configurations, estimating polarimeter errors, and compensating for known error sources. The use of these metrics is illustrated with analyses of two example polarimeters: a dual rotating retarder polarimeter, and a dual variable retarder polarimeter.

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1. Introduction and background

Sample measuring Mueller matrix polarimeters use a set of polarization elements to generate a number of polarization states and then to analyze resulting states which have interacted with a sample of interest. The Mueller matrix of the sample is then reconstructed from the set of measured fluxes. This Mueller matrix is particular to the wavelength, angle of incidence, angular bandwidth, and sample location. In practice error sources couple into the measurement and reducing the accuracy of the reconstruction. Some errors are specific to the calibration and alignment of the polarization elements; for example the waveplate retardance or orientation may be inaccurate. The detection process invariably has noise. This polarimeter optimization process guides the selection of polarization elements and their configurations to enhance stability in the presence of error sources.

Polarimeter optimization has been widely discussed in the literature, most often in terms of light measuring Stokes polarimeters. Azzam et al. first discussed the optimal choice of four analyzers for Stokes polarimeters and demonstrated that the corresponding Stokes vectors define a regular tetrahedron within the Poincaré sphere [1]. Ambirajan and Look presented an optimum rotating retarder Stokes polarimeter configuration by minimizing the condition number of the polarimetric measurement matrix or "instrument matrix" [2, 3]. Sabatke et al. derived the optimal configurations for a rotating retarder Stokes polarimeter using the condition number and related metrics in relation to the Singular Value Decomposition, and showed that there are exactly two ways to inscribe the regular tetrahedron in the Poincaré sphere for this class of polarimeter [4, 5]. Smith generalized these methods to Mueller matrix polarimeters and optimized their configuration using condition number; he demonstrated substantial stability improvement for an over specified polarimeter with more than 16 states [6]. Tyo derived the optimal configuration for a number of Stokes polarimeters, including rotating retarder, variable retarder, multichannel linear, and hyperspectral dual variable retarder polarimeters [7-9]. Savenkov discussed the optimization of a general Mueller matrix polarimeter in terms of condition number for non-depolarizing samples [10]. DeMartino et al. and Garcia-Caurel et al. used condition number to optimize spectroscopic polarimeters using photoelastic modulators and liquid crystal retarders [11-13]. While the condition number provides generalized guidance for the design process, further work is needed to quantify and optimize performance in the presence of specific known error sources. Sabatke et al. used a covariance matrix to optimize a Stokes polarimeter with consideration for random detection noise [5]. Tyo developed a theory to predict the effect of systematic polarimeter errors in Stokes polarimeters, and demonstrated that re-optimization with consideration for systematic error improves stability [14]. Zallat et al. discussed random and systematic errors in relation to Stokes and Mueller matrix polarimeters, with emphasis on Stokes polarimeters having no systematic errors [15]. Broch and Johann developed a method to minimize the effect of random noise in an incomplete Mueller matrix polarimeter by optimization of the orientations of the polarizer and analyzer [16]. Piller et al. considered the effect of systematic error in

polarization element orientation in a dual rotating retarder Mueller matrix polarimeter, and concluded that errors in the compensator elements have stronger effect than errors in the analyzer or polarizer [17]. Uribe-Patarroyo et al. define a system efficiency metric related to modulation intensity variations, and apply this metric to optimize a dual variable retarder Stokes polarimeter in the presence of random noise [18].

This manuscript reviews published optimization and error analysis techniques which have primarily been used in Stokes polarimeter design, and extends these methods for use with sample measuring Mueller matrix polarimeters. As the number of degrees of freedom is significantly larger in Mueller matrix polarimeters (16) than Stokes polarimeters (4), there can be considerable complexity in performing these extensions. In Sections 2 and 3 we review established mathematical formulations and design metrics for Mueller matrix polarimeters. In Section 4 we present a new illustrative example of using the Singular Value Decomposition to evaluate the stability of a Mueller matrix polarimeter in the presence of error. In Section 5 we develop mathematical formulations for error analysis in Mueller matrix polarimeters which are based on previously published covariance matrix methods for Stokes polarimeters. In Section 6 we provide clear illustrative examples of the use of optimization and error analysis techniques by applying these techniques to two common polarimeter configurations. The majority of this illustrative analysis is new, with some previously published material also shown.

2. Formulation for Mueller matrix polarimeters

A sample measuring polarimeter consists of a light source, a polarization state generator (PSG) which delivers polarized light to a sample, the sample, a polarization state analyzer (PSA) which analyzes the light which has interacted with the sample, a detector which measures the flux exiting the analyzer, and associated electronics to control the system and perform calculations. A number of unique combinations of PSG and PSA configurations, called states, comprise the polarimeter; at least 16 linearly independent combinations are required to fully determine all polarization properties of the sample via the Mueller matrix. A common mathematical formulation for polarimeter design [19] represents the N polarimeter states by an $N \times 16$ polarimetric measurement matrix $\mathbf{W}$. The i^{th} polarimeter state $\mathbf{W}_i$ $\{i: 1, 2, \dots, N\}$, is obtained by computing the outer product of the i^{th} generator Stokes vector $\mathbf{G} = \{g_{i0}, g_{i1}, g_{i2}, g_{i3}\}$ and i^{th} analyzer Stokes vector $\mathbf{A} = \{a_{i0}, a_{i1}, a_{i2}, a_{i3}\}$ and flattening the product into a vector,

$$\mathbf{W}_i = [a_{i,0}g_{i,0} \quad a_{i,0}g_{i,1} \quad a_{i,0}g_{i,2} \quad a_{i,0}g_{i,3} \quad a_{i,1}g_{i,0} \quad \cdots \quad a_{i,3}g_{i,3}] . \quad (1)$$

The sequence of measured fluxes for the various states can be arranged into a N -element flux vector $\mathbf{P}$ which is related to the sample Mueller matrix $\mathbf{M}$ by the relation

$$\mathbf{W}\mathbf{M} = \mathbf{P} . \quad (2)$$

$\mathbf{W}$ is typically determined from a model of the polarimeter or from a calibration process. $\mathbf{M}$ is determined from the measurements as the matrix product

$$\mathbf{M} = \mathbf{W}^{-1}\mathbf{P} , \quad (3)$$

where $\mathbf{W}^{-1}$ is the polarimetric data reduction matrix. When $\mathbf{W}$ is not square, the matrix inverse is not unique and the pseudo-inverse

$$\mathbf{W}_p^{-1} = (\mathbf{W}^T\mathbf{W})^{-1}\mathbf{W}^T , \quad (4)$$

provides the minimum RMS solution and so is typically the matrix inverse used [20].

3. Polarimeter design metrics

An examination of the rank, range, and null space of $\mathbf{W}$ provides a useful method to determine whether a proposed polarimeter configuration is suitable to measure an $\mathbf{M}$ of interest. The rank of $\mathbf{W}$ should be greater than or equal to the number of degrees of freedom of $\mathbf{M}$ (16 for a full Mueller matrix). Also, as any polarization state which lies in the null space of $\mathbf{W}$ cannot be measured (a polarimeter is termed “complete” if the null space is empty), all components of $\mathbf{M}$ should lie within the range of $\mathbf{W}$. If $\mathbf{M}$ has components in the polarimeter null space, then the data reduction will find a nearby reconstruction $\mathbf{M}_R \neq \mathbf{M}$ in the range of $\mathbf{W}$. Each row of $\mathbf{W}$ forms one basis vector in the reconstruction of $\mathbf{M}$, i.e. the measured intensity at each polarimeter state is the projection of $\mathbf{M}$ onto the corresponding basis vector. For an effective reconstruction, there should be minimum correlation between basis vectors (i.e. they should be widely distributed over the polarization space), and well balanced in magnitude. For an over specified system with $N > 16$, the basis vectors provide redundant coverage of the polarization space, improving performance in the presence of noise. Basis states may be chosen to lie more densely in directions which provide the most information about $\mathbf{M}$. For example polarimeters to measure stress birefringence are most interested in linear retardance, so the basis states can be selected to improve the signal to noise on those parameters at the expense of diattenuation and depolarization accuracy.

For a general purpose polarimeter, one intended to measure a wide variety of arbitrary $\mathbf{M}$, the polarimetric measurement matrix should be as far from singular as possible, i.e. it should be “well conditioned”. Various linear algebra metrics quantify this distance. The most widely used is κ_p , the condition number based on the L_p norm of the matrix $\mathbf{A}$, defined as [20]

$$\kappa_p(\mathbf{A}) = \|\mathbf{A}\|_p \|\mathbf{A}^{-1}\|_p, \quad (5)$$

where the bars signify the p -norm

$$\|\mathbf{A}\|_p = \sup_{\mathbf{x} \in D(\mathbf{A})} \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p} \quad (\text{matrix } p\text{-norm}), \quad (\|\mathbf{x}\|_p)^p = \sum_i x_i^p \quad (\text{vector } p\text{-norm}), \quad (6)$$

and where $\mathbf{x}$ is a vector, $D(\mathbf{A})$ is the domain of $\mathbf{A}$, and $\sup$ is the supremum (limiting maximum value). Minimization of the condition number of $\mathbf{W}$ is a standard optimization method for polarimeters. Four different condition number definitions are in general use: the L_1 condition number ($p = 1$) based on the maximum absolute column sum; the L_∞ condition number ($p = \infty$) based on the maximum absolute row sum; the L_2 condition number ($p = 2$) based on the Euclidean length of the rows of $\mathbf{A}$; and the Frobenius norm (applicable where $\mathbf{A}$ is square and invertible) based on the determinant of $\mathbf{A}$. Though the various condition numbers differ for a given matrix, they are similarly bounded [20], and so provide equivalent utility. In this article the L_2 condition number is used. The range of the L_2 condition number is from 1 (perfect conditioning) to infinity (singular).

Further insight into the conditioning of $\mathbf{W}$ is obtained from its Singular Value Decomposition (SVD) which was introduced to polarimeter design by Tyo [14] and Sabatke et al. [5]. The SVD factors any $N \times K$ matrix $\mathbf{A}$ as

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T = \mathbf{U} \begin{pmatrix} \mu_0 & & & & \\ & \mu_1 & & & \\ & & \ddots & & \\ & & & \mu_{K-1} & \\ & & & & \mu_K \\ 0 & 0 & \cdots & 0 & 0 \\ & \vdots & & & \end{pmatrix} \mathbf{V}^T, \quad (7)$$

where $\mathbf{U}$ and $\mathbf{V}$ are $N \times N$ and $K \times K$ unitary matrices, and $\mathbf{D}$ is an $N \times K$ diagonal matrix. The diagonal elements μ_k are the singular values. The rank of $\mathbf{W}$ is given by the number of nonzero singular values; for a full rank polarimetric measurement matrix the rank is 16. Those columns of $\mathbf{U}$ associated with nonzero singular values form an orthonormal basis for the range of $\mathbf{W}$; those columns of $\mathbf{V}$ associated with zero-valued singular values form an orthonormal basis for the null space of $\mathbf{W}$. The columns of $\mathbf{V}$ associated with nonzero singular values form an orthonormal basis which spans the full vector space of $\mathbf{W}$ and thus reconstructs $\mathbf{M}$. Each singular value gives the relative strength of the corresponding vector in this basis set, and the columns of $\mathbf{U}$ form a mapping from the $\mathbf{V}$ basis set back to the original basis set of $\mathbf{W}$. Further, since

$$\mathbf{P} = \mathbf{W}\mathbf{M} = \mathbf{U} \begin{pmatrix} \mu_1 & 0 & 0 & 0 \\ 0 & \mu_2 & 0 & 0 \\ & & \ddots & \\ 0 & 0 & 0 & \mu_{16} \\ & \vdots & & \end{pmatrix} \mathbf{V}^T \mathbf{M}, \quad (8)$$

the rows of $\mathbf{U}$ corresponding to zero-valued singular values describe sets of flux measurements which are not generated by any Mueller matrix, so their presence in a polarimetric measurement can only be due to noise. Based on this interpretation, any basis vector in $\mathbf{V}$ which is associated with a relatively small singular value is near the null space and likely has little information content; such singular values predominantly amplify noise into the reconstruction of $\mathbf{M}$. Error sources which produce projections (intensity vectors) which are similar to the intensity vectors generated by the basis vectors in $\mathbf{V}$ (particularly those which correspond to large singular values) will couple strongly into the reconstruction of $\mathbf{M}$. The L_2 condition number is equal to the ratio of the largest to smallest singular values [21], and thus minimizing the condition number is equivalent to equalizing, to the extent possible, the range of singular values so that the basis vectors have wide distribution and similar weight.

A geometrical interpretation relating the stability of a Stokes polarimeter to the trajectory on the Poincaré sphere was first proposed by Azzam et al. [1] and later extended by Ambirajan and Look [2,3], Sabatke et al. [4], and Tyo [7]. For a four measurement Stokes polarimeter, the Stokes vectors representing each of the four PSA states, when plotted on the Poincaré sphere, define a tetrahedron which is generally irregular. The volume of the tetrahedron is proportional to the determinant of $\mathbf{W}$, and is maximized when the vertices form a regular tetrahedron. In this case the maximum distance from a vertex to any point on the sphere is minimized, and the condition number is also minimum. The selection of PSA polarization elements may thus be limited by considering only those combinations providing appropriate degrees of freedom such that a regular tetrahedron may be inscribed in the Poincaré sphere trajectory. For a rotating linear retarder in front of a fixed linear polarizer (one degree of freedom), Sabatke et al. demonstrated a “figure eight” trajectory with varying

aspect ratio dependent on retardance [4]. An optimal trajectory enclosing a regular tetrahedron was identified for exactly two retardances. Tyo showed that for two degrees of freedom (such as a dual variable retarder Stokes polarimeter) the trajectory forms a two-dimensional surface on the Poincaré sphere, giving a continuum of optimum configurations in which a regular tetrahedron may be inscribed [7].

A geometrical interpretation for a sample measuring polarimeter is obtained by considering that $\mathbf{W}$ is formed as the outer product of the PSG and PSA Stokes vectors. The condition number of $\mathbf{W}$ is then related to the condition numbers of the generator and analyzer as

$$\kappa(\mathbf{W}) = \kappa(\mathbf{G}^T \otimes \mathbf{A}) = \kappa(\mathbf{G}^T) \kappa(\mathbf{A}) = \kappa(\mathbf{G}) \kappa(\mathbf{A}) . \quad (9)$$

Therefore the PSG and PSA (often identical) should both be individually optimized in terms of Poincaré sphere trajectory.

4. Singular Value Decomposition examples

As an example of an optimum polarimeter, consider a Mueller matrix polarimeter which uses four generator states $\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{V}_4$, located at the vertices of a regular tetrahedron on the Poincaré sphere,

$$\mathbf{V}_1 = \{1, 1, 0, 0\}, \mathbf{V}_2 = \left\{1, -\frac{1}{3}, \frac{2\sqrt{2}}{3}, 0\right\}, \mathbf{V}_3 = \left\{1, -\frac{1}{3}, -\frac{\sqrt{2}}{3}, \sqrt{\frac{2}{3}}\right\}, \mathbf{V}_4 = \left\{1, -\frac{1}{3}, -\frac{\sqrt{2}}{3}, -\sqrt{\frac{2}{3}}\right\} . \quad (10)$$

The analyzer states are also chosen as $\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{V}_4$. Sixteen measurements are acquired at each of the combinations of generator and analyzer. This is one member of the set of 16-measurement Mueller matrix polarimeters with minimum condition number, so it can be considered an optimum configuration. The corresponding 16 singular values are

$$\left\{4, \frac{4\sqrt{3}}{3}, \frac{4\sqrt{3}}{3}, \frac{4\sqrt{3}}{3}, \frac{4\sqrt{3}}{3}, \frac{4\sqrt{3}}{3}, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}\right\} , \quad (11)$$

and the condition number, equal to the quotient of the first and last singular values, is 3. Each of the sixteen columns of $\mathbf{U}$ represents a different orthogonal component used to reconstruct a measured Mueller matrix. In the presence of white noise, the Mueller matrix component corresponding to the first column will be measured with the highest signal to noise, about $\sqrt{3}$ times better than the next six components (columns) from $\mathbf{U}$, and about three times better than the last ten Mueller matrix components.

As an example of a polarimeter with a nearly singular polarimetric measurement matrix, the second row of $\mathbf{W}$,

$$\{1, -0.333, 0.943, 0, 1, -0.333, 0.943, 0, 0, 0, 0, 0, 0, 0, 0\} , \quad (12)$$

will be replaced with a vector

$$\{1, 1, 0.0005, 0, 1, 1, 0.0005, 0, 0, 0, 0, 0, 0, 0, 0\} . \quad (13)$$

This vector is nearly equal to the first row of $\mathbf{W}$,

$$\{1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0\} , \quad (14)$$

so that these two rows are nearly linearly dependent. Examining the singular values,

$$\{4.0557, 2.7273, 2.3094, 2.3094, 2.3094, 2.3094, 2.0221, 1.4988, 1.3333, 1.3333, 1.3333, 1.3333, 1.3333, 1.3333, 1.3333, 0.0004\} , \quad (15)$$

we see that the last singular value is close to zero and the condition number is about 10^4 . Whenever the measured flux contains the pattern corresponding to the last row of $\mathbf{V}^T$, this component will be amplified by about 10^4 during the data reduction relative to the other 15 components of the Mueller matrix and will dominate the measurement. In the presence of random noise, the measured Mueller matrix will be close to the 16th column of $\mathbf{U}$ (partitioned into a 4x4 “Mueller matrix”), and so the measurement will be inaccurate.

In summary, components corresponding to very small singular values are greatly amplified in the matrix inverse and can overwhelm the remainder of the Mueller matrix in the polarimetric data reduction.

5. Error analysis

The metrics presented in the previous section guide polarimeter design toward higher stability. The goals of the error analysis method presented in this section are as follows: (1) to quantify the accuracy in reconstructing $\mathbf{M}$, (2) to identify those error sources with largest contributions, (3) to compensate $\mathbf{W}$ for known errors, (4) and to optimize the polarimeter design with consideration for error.

When operating a polarimeter, $\mathbf{W}$ is not known exactly and may have changed since calibration. For example, a rotating retarder may have inconsistent orientation, rays may take different paths through the polarimeter for different samples, or the spectral distribution of the measured light may vary. Measurement error is also present, due to detector noise and source fluctuation. Equation 2 may be modified to include these effects as follows,

$$(\mathbf{W} + \delta\mathbf{W})\mathbf{M} + \delta\mathbf{P} = \mathbf{P}_M , \quad (16)$$

where $\delta\mathbf{W}$ is a $N \times 16$ matrix representing the difference between the actual and calibrated $\mathbf{W}$, $\delta\mathbf{P}$ is a $N \times 1$ vector representing intensity measurement error, and $\mathbf{P}_M$ is the $N \times 1$ vector of intensities measured in the presence of error. $\mathbf{M}_R$, the polarimeter’s estimate of $\mathbf{M}$, is then calculated using the calibration data as

$$\begin{aligned} \mathbf{M}_R &= \mathbf{W}_p^{-1} \mathbf{P}_M \\ &= \mathbf{W}_p^{-1} \cdot [(\mathbf{W} + \delta\mathbf{W})\mathbf{M} + \delta\mathbf{P}] \\ &= \mathbf{M} + \delta\mathbf{M} = \mathbf{M} + \mathbf{W}_p^{-1} \cdot [\delta\mathbf{W}\mathbf{M} + \delta\mathbf{P}] , \end{aligned} \quad (17)$$

where $\delta\mathbf{M}$ is the difference between the measured $\mathbf{M}_R$ and the actual $\mathbf{M}$. There are two error terms in $\delta\mathbf{M}$, one dependent on $\delta\mathbf{W}$ and $\mathbf{M}$, and the other on $\delta\mathbf{P}$.

Small errors may be represented by a first order Taylor expansion. The error for the j^{th} component of the i^{th} polarimeter state, having R variables x_r which may be subject to error (such as a retardance magnitude), each with nominal value ϕ_r , and error magnitude δ_r , is then

$$\delta\mathbf{W}_{ij} \approx \sum_{r=1}^R \delta_{rij} \left. \frac{\partial \mathbf{W}_{ij}}{\partial x_r} \right|_{x_r = \phi_r} . \quad (18)$$

The error in $\mathbf{P}$ is assumed independent of the polarimeter elements, and is given by

$$\delta\mathbf{P}_i = \epsilon_i , \quad (19)$$

where ε_i is the error in the i^{th} intensity measurement. The error in reconstructing each of the $k = 1 \dots 16$ elements of $\mathbf{M}$ in terms of the errors in the instrument and detection process is then

$$\delta \mathbf{M}_k = \sum_{i=1}^N \mathbf{W}_{ki}^{-1} \left[\sum_{j=1}^{16} \sum_{r=1}^R \delta_{rij} \frac{\partial \mathbf{W}_{ij}}{\partial x_r} \right]_{x_r = \phi_r} \mathbf{M}_j + \sum_{i=1}^N \mathbf{W}_{ki}^{-1} \varepsilon_i \quad k: 1 \dots 16. \quad (20)$$

The mean (expectation) and standard deviation (SD) of the error ($\langle \delta \mathbf{M} \rangle$ and $\text{SD}(\delta \mathbf{M})$) may be estimated when the polarimeter state variables and the statistics of the error sources are approximately known.

The error due to a known systematic (nonzero mean error) source may be compensated by estimating $\delta \mathbf{W}$ (for example using ideal Mueller matrices to model the polarimeter) and then forming a new polarimetric measurement matrix $\mathbf{W}_q = \mathbf{W} + \langle \delta \mathbf{W} \rangle$. For example, when using a liquid crystal retarder with a known profile of retardance magnitude as a function of temperature, a new $\mathbf{W}_q$ may be recalculated at every use given the ambient temperature.

A covariance matrix may be used to optimize a polarimeter in the presence of known error. This method has been applied to random measurement noise in Stokes polarimetry by Sabatke et al. [5], and to random instrument noise in Stokes polarimetry by Tyo [14]. The covariance matrix is a symmetric matrix which describes the correlation between random variables which have been centered about their means. For Mueller matrix polarimetry, the 16x16 covariance matrix, $\mathbf{C}_M$, is given by

$$\mathbf{C}_{M,jk} = \langle \delta \mathbf{M}_j \delta \mathbf{M}_k \rangle - \langle \delta \mathbf{M}_j \rangle \langle \delta \mathbf{M}_k \rangle \quad j, k : 1 - 16. \quad (21)$$

To derive $\mathbf{C}_M$, consider the total error $\delta \mathbf{E}$ for the i^{th} polarimeter state:

$$\delta \mathbf{E}_i = \sum_{j=1}^{16} \delta \mathbf{W}_{ij} \mathbf{M}_j + \delta \mathbf{P}_i. \quad (22)$$

If all error sources are assumed to be uncorrelated with the same statistical properties among all polarimeter states, then the error may be simplified to a sum of terms, each involving the polarimeter errors δ_r and detection error ε :

$$\delta \mathbf{E}_i = \sum_{j=1}^{16} \sum_{r=1}^R a_{rji} \delta_r m_j + b_i \varepsilon, \quad (23)$$

where the a 's and b 's are constant coefficients derived from $\delta \mathbf{W}$ (typically many are zero) and the m 's are the 16 elements of $\mathbf{M}$ when flattened from a 4x4 matrix to a 16x1 matrix.

An error covariance matrix $\mathbf{C}_E$ based on $\delta \mathbf{E}$ is given by

$$\mathbf{C}_{E,jk} = \langle \delta \mathbf{E}_j \delta \mathbf{E}_k \rangle - \langle \delta \mathbf{E}_j \rangle \langle \delta \mathbf{E}_k \rangle \quad j, k : 1 - 16. \quad (24)$$

The only nonzero terms of $\mathbf{C}_E$ are on the diagonals since error sources are assumed to be uncorrelated. These diagonal terms have the form

$$\mathbf{C}_{\mathbf{E},ii} = \sum_{k=1}^{16} \sum_{j=1}^{16} m_j m_k \left[\sum_{r=1}^R c_{rjk} \sigma_r^2 + d_i \sigma_{\varepsilon}^2 \right], \quad (25)$$

where the c 's and d 's are constant coefficients derived from $\delta\mathbf{E}$ (typically many are zero), the σ_r^2 's are the standard deviations of the polarimeter error sources, and σ_{ε}^2 is the standard deviation of the detection process error. It is straightforward to show that $\mathbf{C}_{\mathbf{M}}$ may be expressed as

$$\mathbf{C}_{\mathbf{M}} = \mathbf{W}^{-1} \mathbf{C}_{\mathbf{E}} (\mathbf{W}^{-1})^T. \quad (26)$$

The on-diagonal elements of $\mathbf{C}_{\mathbf{M}}$ represent gain factors in the propagation of error into $\mathbf{M}_{\mathbf{R}}$. One possible error metric, EM , is the sum of the diagonal elements,

$$EM = \sum_i \mathbf{C}_{\mathbf{M},ii}. \quad (27)$$

EM is a function of the polarimeter configuration, the number of states, the Mueller matrix of the sample, and the statistical properties of the error sources. Minimization of EM with respect to a polarimeter variable may be used to compute the variable's optimal value in the presence of known error.

6. Illustrative examples

To illustrate optimization and error analysis techniques, two common Mueller matrix polarimeter configurations are analyzed: Dual Rotating Retarder (DRR) and Dual Variable Retarder (DVR). These configurations are shown schematically in Fig. 1; the polarization elements are summarized in Table 1. Table 2 lists typical errors with magnitudes taken from recent polarimeters constructed in our laboratory. Though additional error sources are typically present in DRR and DVR polarimeter designs (for example error in DRR retarder retardance or DVR retarder orientation), we limited the analyzed error sources to those in Table 2 for simplicity in illustrating the described analysis methods. All error calculations should be considered as due to a positive error (for example an over-rotation of an element); an error of equal magnitude and opposite sign is implied for the case of a negative error.

The DRR polarimeter PSG (or equivalently the PSA) with one variable parameter (retarder rotational increments $\Delta\phi_g$) traces a figure-eight trajectory on the Poincaré sphere. The aspect ratio of the figure-eight is adjustable via the choice of the fixed retardance γ . For the optimal choice of γ (found for example using a condition number analysis), a regular tetrahedron may be inscribed within the trajectory as shown in Fig. 2(a). The DVR polarimeter PSG (or equivalently the PSA) with two variable parameters (retardances of the two variable retarders γ_{g1} and γ_{g2}) traces a surface on the Poincaré sphere. The extent of the surface is adjustable via the choice of the fixed retarder orientations ϕ_1 and ϕ_2 . Many choices define a surface within which a regular tetrahedron may be inscribed as shown in Fig. 2(b).

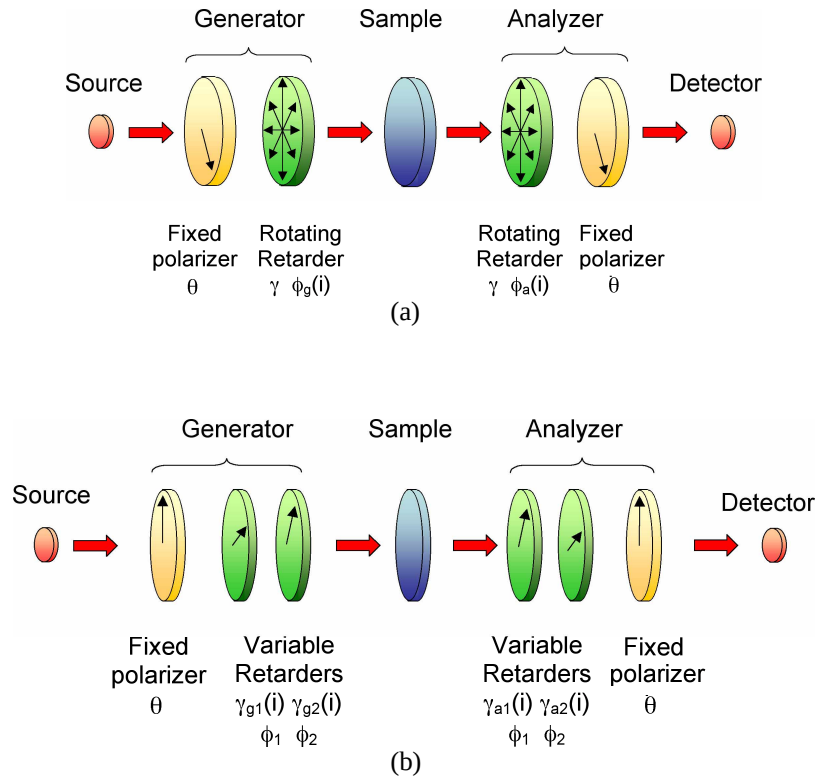


Fig. 1. Polarimeter configurations used in illustrative examples. (a) Dual Rotating Retarder (DRR) polarimeter. Components include fixed polarizers at orientation θ , and rotating retarders with retardance γ and orientations ϕ_g and ϕ_a . (b) Dual Variable Retarder (DVR) polarimeter. Components include fixed polarizers with orientation θ , outer retarders with fixed orientation ϕ_1 and retardances γ_{g1} and γ_{a1} , and inner retarders with fixed orientation ϕ_2 and retardances γ_{g2} and γ_{a2} .

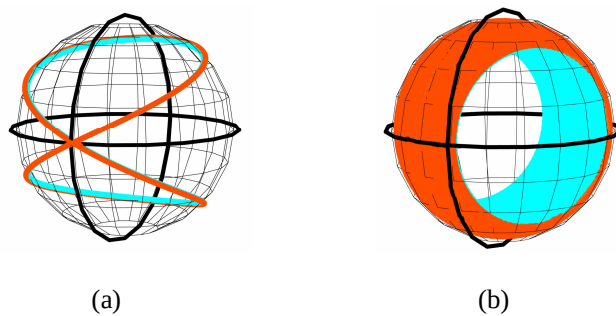


Fig. 2. Trajectories on the Poincaré sphere. (a) DRR polarimeter with one variable traces a figure-eight trajectory; trajectory shown is for $\gamma = 127^\circ$. (b) DVR polarimeter with two variables traces a surface; surface shown is for $\phi_1 = 27.4^\circ$, $\phi_2 = 72.4^\circ$.

Table 1. Summary of polarimeter polarization components

Polarimeter	Fixed parameters in generator and analyzer	Variable parameters in generator and analyzer
DRR	Retarder retardance γ Polarizer orientation θ	Retarder rotation increments $\Delta\phi_g(i)$, $\Delta\phi_a(i)$
DVR	Retarder 1 orientation ϕ_1 Retarder 2 orientation ϕ_2 Polarizer orientation θ	Retarder 1 retardance $\gamma_{g1}(i)$, $\gamma_{a1}(i)$ Retarder 2 retardance $\gamma_{g2}(i)$, $\gamma_{a2}(i)$

Table 2. Summary of polarimeter error sources

Polarimeter	Systematic error sources	Worst Case Mean Error	Random error sources	Worst Case SD Error
DRR	None	N/A	Retarder position Detector noise	$\sigma_{g,a}^2 = 0.5^\circ$ $\sigma_p^2 = 0.01$
DVR	Retardance drift	$\langle \delta_{1,2} \rangle = 2.5\%$	Detector noise	$\sigma_p^2 = 0.01$

To evaluate the condition number metric, an instrument matrix $\mathbf{W}$ is assembled as a function of all polarimeter variables such as retardances and orientations. The condition number of $\mathbf{W}$ as a function of the variables is then calculated. The $\mathbf{W}$ having rank = 16 and lowest condition number gives the optimal values for the variables. The DRR polarimeter has been analyzed in terms of condition number for $N = 16$ and $N = 30$ by Smith [6]. For the $N = 16$ case an optimal retardance of $\gamma = 127^\circ$ and optimal angular increments of $\Delta\theta_g = 34^\circ$ and $\Delta\theta_a = 26^\circ$ were found. Multiple solutions were found for the $N = 30$ case. The minimum L_2 condition number is 16.7 for $N = 16$, and 3.48 for $N = 30$. A condition number analysis to determine the optimal waveplate retardance is shown in Fig. 3. A condition number analysis to determine the optimal angular increments is shown in Fig. 4(a) for $N=16$ (compare to Smith [6] Fig. 3); and in Fig. 4(b) for $N=30$ (compare to Smith [6] Fig. 4). For our plots we have chosen a nonlinear color scale to emphasize the low condition number solutions.

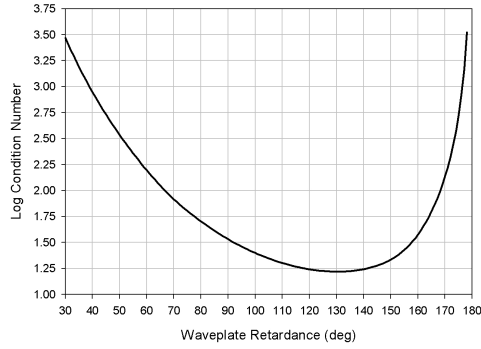


Fig. 3. Base 10 log of the condition number as a function of waveplate retardance γ for the DRR polarimeter. The optimum solution is a retardance of 127° ; condition number increases significantly with distance from the optimum solution.

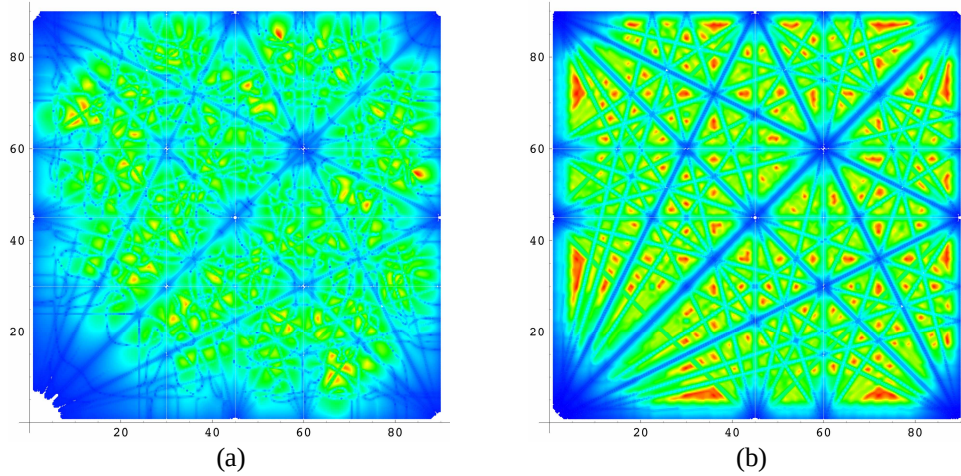


Fig. 4. Condition number as a function of rotating retarder increment for the DRR polarimeter. Horizontal and vertical axes are the rotational increments of the two retarders ($\Delta\theta_g$ and $\Delta\theta_a$) in degrees. Blue indicates a high (undesirable) condition number; red indicates a low condition number. Color scale is nonlinear to emphasize the low condition number solutions. (a) For $N = 16$, the best solutions are highly localized; one example is $\Delta\theta_g = 34^\circ$ and $\Delta\theta_a = 26^\circ$. (b) For $N = 30$ there are many extended regions with good solutions.

The DVR polarimeter has been analyzed in terms of condition number by De Martino et al. [12] for the case of the four retardances γ_{g1} , γ_{g2} , γ_{a1} , and γ_{a2} selected from permutations of a pair of retardances (γ_1, γ_2) . Optimal retardances $\gamma_1 = 225^\circ$, $\gamma_2 = 45^\circ$, and orientations $\phi_1 = 27.4^\circ$, and $\phi_2 = 72.4^\circ$ were found. The minimum L2 condition number is 3, lower than for the DRR polarimeter, indicating better performance in the presence of error. A condition number optimization of orientations when retardances are fixed is shown in Fig. 5(a); a condition number optimization with respect to retardances when orientations are fixed is shown in Fig. 5(b). As compared to the DRR polarimeter, the low condition number solutions are clustered near fewer optimal values. Larger cluster size indicates a lower sensitivity to uncertainty in the angles and retardances.

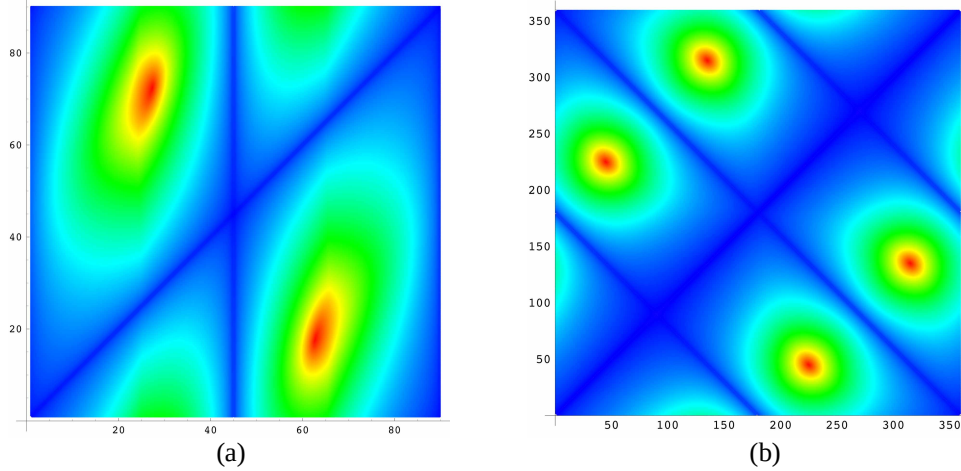


Fig. 5. Condition number as a function of polarimeter variables for the DVR polarimeter, for $N = 16$. Blue indicates a high (undesirable) condition number, and red indicates a low condition number. Color scale is nonlinear to emphasize the low condition number solutions. (a) Optimization of retarder angular orientations ϕ_1 and ϕ_2 assuming retardances of $\gamma_1 = 225^\circ$, $\gamma_2 = 45^\circ$. Horizontal and vertical axes are the angular orientations of the two retarders in degrees. (b) Optimization of retarder retardances γ_1 and γ_2 for the optimum orientations of $\phi_1 = 27.4^\circ$, $\phi_2 = 72.4^\circ$. Horizontal and vertical axes are the retardances of the two retarders in degrees.

The singular values (SVs) for each polarimeter as optimized using condition number are shown in Fig. 6. For each polarimeter there are 16 nonzero SVs, indicating a rank 16 instrument matrix. The SV profiles for the $N = 30$ DRR and DVR polarimeters are more optimal, as they exhibit low slope for singular values 2-7 and 8-16, indicating more equal weighting of the basis vectors.

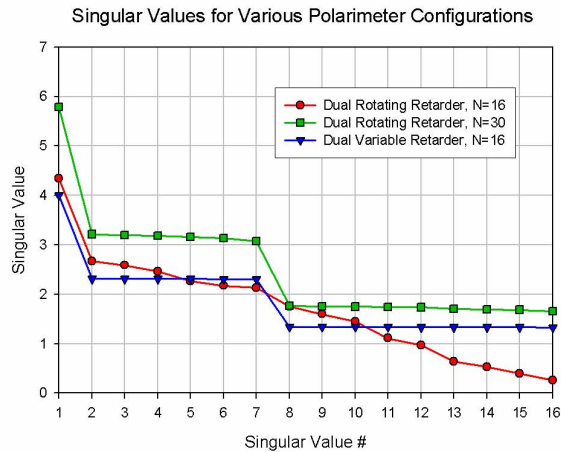


Fig. 6. Singular values for the example polarimeter configurations. Lower slope and absolute singular value range indicates a more stable configuration.

The predicted error in reconstruction of $\mathbf{M}$ was calculated for two samples: a linear polarizer, rotated from 0° to 90° in steps of 15° ; and a quarter wave plate, rotated from 0° to 90° in steps of 15° , assuming the errors listed in Table 2. Predicted error is shown in Fig. 7. For the first sample, worst case mean and standard deviation of diattenuation magnitude and orientation are calculated as a function of polarizer orientation. For the second sample, worst

case mean and standard deviation of retardance magnitude and orientation are calculated as a function of retarder orientation. Results for the DRR polarimeter show zero mean error since there are no systematic error sources. The standard deviation is significant due to the random error in retarder orientation, and has a strong dependence on orientation. The DVR polarimeter has measurable mean error due to the retardance error of the variable retarders, with clear dependence on orientation. The standard deviation is small since the only random error source (detector noise) has small value.

The error metric EM for the DRR polarimeter was used to determine the optimal retardance γ in the presence of error as listed in Table 2. A plot of EM, Fig. 8(a), shows that γ depends on $\mathbf{M}$; for example, $\gamma = 117^\circ$ for a linear polarizer sample at 45° orientation, and $\gamma = 128^\circ$ for a quarter wave plate sample at 45° orientation. A comparison of EM for $N = 16$ and $N = 30$ for a horizontal linear polarizer sample, Fig. 8(b), shows a $1/N^2$ improvement in EM and a slight shift in γ , from 122° to 125° . Thus taking additional measurements improves immunity to error sources and pulls the optimized retardance back towards the value found using the condition number metric (127°).

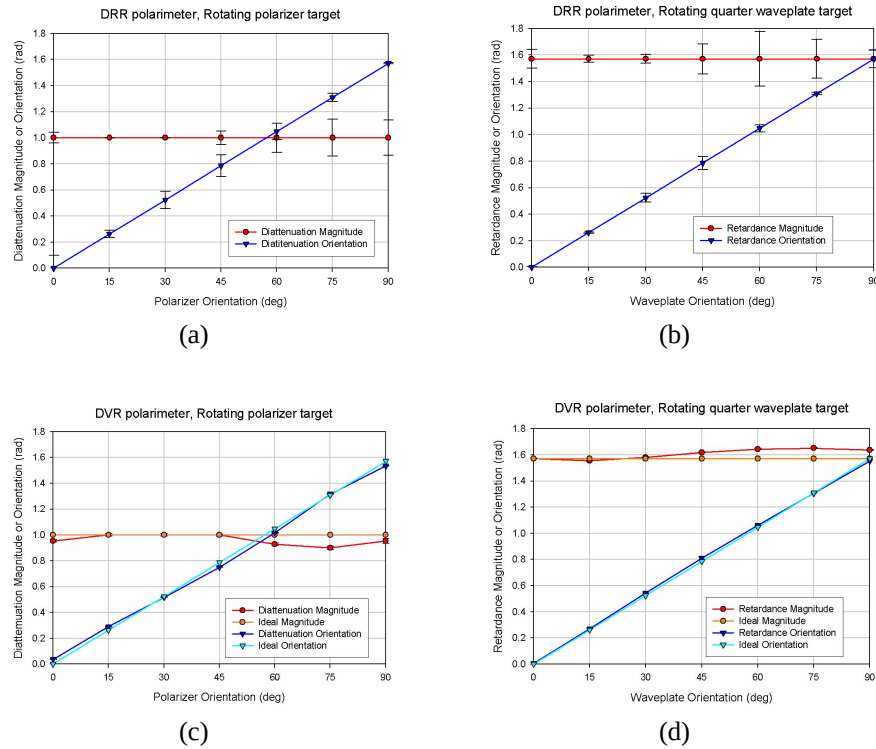


Fig. 7. Calculated mean and SD error for DRR (a,b) and DVR (c,d) polarimeters. (a), (c) Sample is a linear polarizer rotated from 0° to 90° in 15° increments; (b), (d) Sample is a quarter wave plate rotated from 0° to 90° in 15° increments. Solid red and blue lines indicate worst case error from systematic sources; error bars indicate worst case error from random sources.

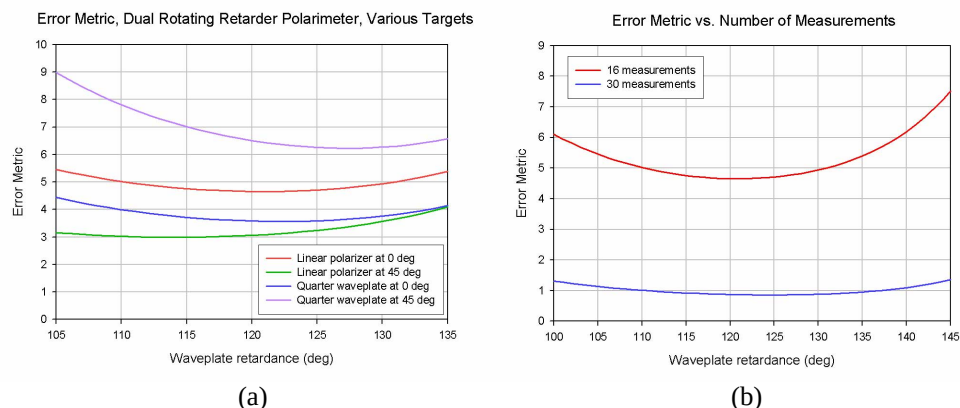


Fig. 8. EM metric to optimize retardance for the DRR polarimeter when information is known about the error source and/or sample to be measured. (a) The dependence of optimal retardance on $\mathbf{M}$; and (b) The dependence of optimal retardance on N for a horizontal linear polarizer.

7. Conclusions

Techniques for optimizing sample-measuring Mueller matrix polarimeters were presented which maximize accuracy and stability in the sample Mueller matrix reconstruction. The most important design requirements are the following: (1) achieving a rank 16 polarimeter, so that no polarization properties lie in the null space and thus introduce error into the Mueller matrix reconstruction; and (2) minimizing the condition number of the instrument matrix to ensure stability in the presence of error sources. Poincaré sphere trajectories may be used as a visualization tool to limit the potentially large solution space. The Singular Value Decomposition provides insight into the stability of the instrument matrix via the balancing of the basis vectors. An error analysis based on a Taylor expansion of the instrument matrix may be used to assess the individual and/or combined effects of the various error sources, to estimate the magnitude of the error in the reconstruction, and to provide compensation when error sources are predictable and well characterized. The tasks of appropriate selection of polarization elements and comparison of different polarimeter configurations have been illustrated with two examples.

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